



BHARATIYA VIDYA BHAVAN'S

SARDAR PATEL INSTITUTE OF TECHNOLOGY

MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400 058, India
(Autonomous College Affiliated to University of Mumbai)

Makeup Examination *July 2019*

Max. Marks: 60

Class: FYMCA

Course Code: MCA21

Subject: Operating System

Duration: 3 hr

Semester: II

Date:

Time:

- Instructions: (1) All questions are compulsory.
(2) Use of scientific calculator is allowed.
(3) Assume any necessary data but justify the same.

Q. No.	Questions	Max. Marks	CO																
Q1	a) Compare different system software based on their working OR	5	CO1																
	a) Classify different types of OS based on their functionality	5	CO1																
Q2	a) Consider below table of four processes with their arrival and burst time	6	CO2																
	<table><tr><th>Process</th><th>AT</th><th>BT</th></tr><tr><td>P1</td><td>0</td><td>8</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>2</td><td>9</td></tr><tr><td>P4</td><td>3</td><td>5</td></tr></table>	Process	AT	BT	P1	0	8	P2	1	4	P3	2	9	P4	3	5			
	Process	AT	BT																
	P1	0	8																
P2	1	4																	
P3	2	9																	
P4	3	5																	
	Find Average waiting time for 1.Preemptive SJF 2.FCFS OR																		
	a) Consider the system with 3 processes P1, P2, P3. Analyze whether the given processes are schedulable or not using Rate Monotonic Scheduling Algorithm (1 is the highest Priority)	6	CO2																
	<table><tr><th>Process</th><th>Service Time</th><th>Time Period(Repeat frequency)</th><th>Priority</th></tr><tr><td>P1</td><td>12</td><td>50</td><td>3</td></tr><tr><td>P2</td><td>10</td><td>40</td><td>2</td></tr><tr><td>P3</td><td>10</td><td>30</td><td>1</td></tr></table>	Process	Service Time	Time Period(Repeat frequency)	Priority	P1	12	50	3	P2	10	40	2	P3	10	30	1		
Process	Service Time	Time Period(Repeat frequency)	Priority																
P1	12	50	3																
P2	10	40	2																
P3	10	30	1																
	b) Compare Multilevel queue scheduling with multilevel feedback queue scheduling	4	CO2																



BHARATIYA VIDYA BHAVAN'S

SARDAR PATEL INSTITUTE OF TECHNOLOGY

MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400 058, India
(Autonomous College Affiliated to University of Mumbai)

Q3	a) Define Semaphore State how does semaphores solve problem of 1. Critical section 2. Order of execution among the processes(Give example) 3. Resource management(Give example) OR a) Consider a system with 4 processes and 3 resource types: A(14 instances),B(10 instances) and C(10 instances) <table><tr><th rowspan="2">Process</th><th colspan="2">A</th><th colspan="2">B</th><th colspan="2">C</th></tr><tr><th>Allocation</th><th>Max</th><th>Allocatio</th><th>Max</th><th>Allocation</th><th>Max</th></tr><tr><td>P1</td><td>2</td><td>4</td><td>1</td><td>2</td><td>1</td><td>3</td></tr><tr><td>P2</td><td>2</td><td>7</td><td>3</td><td>3</td><td>1</td><td>10</td></tr><tr><td>P3</td><td>1</td><td>5</td><td>1</td><td>2</td><td>4</td><td>13</td></tr><tr><td>P4</td><td>1</td><td>10</td><td>4</td><td>5</td><td>2</td><td>4</td></tr></table> Using Banker's Algorithm 1. Find the content of Need Matrix 2. Is the system in safe state? If so, Give safe sequence. b) State the necessary conditions to solve critical section problem	Process	A		B		C		Allocation	Max	Allocatio	Max	Allocation	Max	P1	2	4	1	2	1	3	P2	2	7	3	3	1	10	P3	1	5	1	2	4	13	P4	1	10	4	5	2	4	1 3 3 3 10 5	CO2 CO2 CO2
Process	A		B		C																																							
	Allocation	Max	Allocatio	Max	Allocation	Max																																						
P1	2	4	1	2	1	3																																						
P2	2	7	3	3	1	10																																						
P3	1	5	1	2	4	13																																						
P4	1	10	4	5	2	4																																						
Q4	a) Consider the requests from processes in given order 300K, 25K, 125K and 50K. Let there be two blocks of memory available of size 150K followed by a block size 350K. Which of the partition allocation schemes can satisfy above requests? b)State how does TLB overcomes drawbacks of paging c) What is Page Fault? Consider a reference string: 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. the number of frames in the memory is 3. Find out the number of page faults respective to: 1. Optimal Page Replacement Algorithm 2. FIFO Page Replacement Algorithm 3. LRU Page Replacement Algorithm OR	5 5 10	CO3 CO3 CO3																																									



BHARATIYA VIDYA BHAVAN'S

SARDAR PATEL INSTITUTE OF TECHNOLOGY
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI – 400 058, India
(Autonomous College Affiliated to University of Mumbai)

	c) Consider the following disk request sequence for a disk with 100 tracks 98, 137, 122, 183, 14, 133, 65, 78 .Head pointer starting at 54 and moving in left direction. Find the number of head movements in cylinders using: 1. FCFS 2. SSTF 3. SCAN 4. C-SCAN 5. LOOK	10	CO3
Q5	a) Compare different disk free space management techniques	6	CO4
	b) State the goals, principles and domain of protection	4	CO4